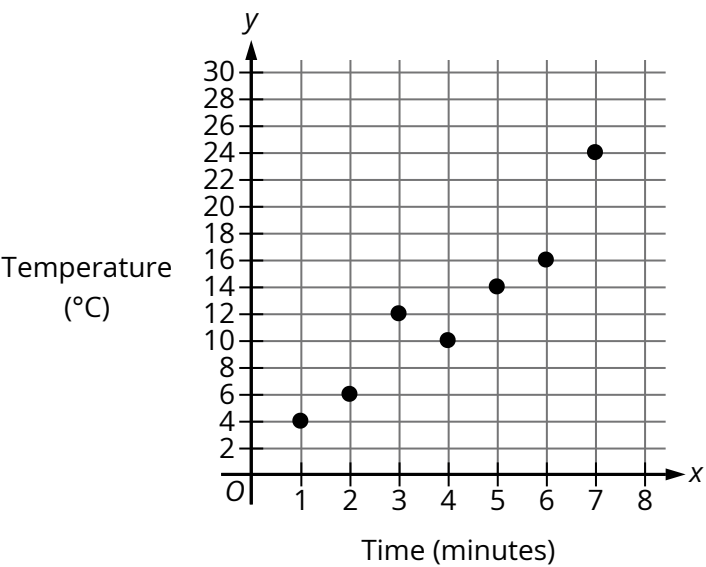


Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Problem-Solving and Data Analysis	Two-variable data: Models and scatterplots	Medium

Question

During a study, the temperature, **in degrees Celsius ($^{\circ}\text{C}$)**, of the air in a chamber was recorded to the nearest integer at certain times. The scatterplot shows the recorded temperature y , **in $^{\circ}\text{C}$** , of the air in the chamber x minutes after the start of the study.



What was the average rate of change, **in $^{\circ}\text{C}$ per minute**, of the recorded temperature of the air in the chamber from $x = 5$ to $x = 7$?

Correct Answer: 5

Rationale

The correct answer is **5**. For the graph shown, x represents time, in minutes, and y represents temperature, in degrees Celsius ($^{\circ}\text{C}$). Therefore, the average rate of change, in $^{\circ}\text{C}$ per minute, of the recorded temperature of the air in the chamber between two x -values is the difference in the corresponding y -values divided by the difference in the x -values. The graph shows that at $x = 5$, the corresponding y -value is **14**. The graph also shows that at $x = 7$, the corresponding y -value is **24**. It follows that the average rate of change, in $^{\circ}\text{C}$ per minute, from $x = 5$ to $x = 7$ is $\frac{24-14}{7-5}$, which is equivalent to $\frac{10}{2}$, or **5**.